$$A(0) = E^*[e^{-rT} K D \cdot I\{V(T) \geq D\}]$$

$$= e^{-rT} K D \; E^*[I\{V(T) \geq D\}]$$

$$= e^{-rT} K D \; P^*(V(T) \geq D)$$

$$= e^{-rT} K D \left(1 - P^*(V(T) \leq D)\right)$$

$$= \text{- - - - (- ———)}$$

$$I\{V(T) \geq D\} = \begin{cases} 1, & \text{if } V(T) \geq D \\ 0, & \text{if } V(T) < D \end{cases}$$

$$E^*[I\{V(T) \geq D\}]$$

$$= 1 \times P(V(T) \geq D) + \underbrace{0 \times P(V(T) < D)}_{=0}$$

$$= P(V(T) \geq D$$

$V_T = V(T)$ ✓✓✓

$$P^*(V_T \le D) = P^*\left(\frac{V_T}{V_0} \le \frac{D}{V_0}\right)$$

$$= P^*\left(\ln\left(\frac{V_T}{V_0}\right) \le \ln\left(\frac{D}{V_0}\right)\right)$$

$$= P^*\left(\ln\left(\frac{V_T}{V_0}\right) - \left(r - \tfrac{1}{2}\sigma^2\right)T \le \ln\left(\frac{D}{V_0}\right) - \left(r - \tfrac{1}{2}\sigma^2\right)T\right)$$

$$= P^*\left(\frac{\ln\left(\frac{V_T}{V_0}\right) - \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}} \le \frac{\ln\left(\frac{D}{V_0}\right) - \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}\right)$$

$$= P^*\left(Z \le \frac{\ln\left(\frac{D}{V_0}\right) - \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}\right)$$

$$= \Phi\left(\frac{\ln\left(\frac{D}{V_0}\right) - \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}\right)$$

$$dV_t = \gamma V_t dt + \sigma V_t dW_t$$